

33 Carboxylic acids and derivatives

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Acyl chlorides, acid anhydrides, esters and amides vary in ease of nucleophilic acyl substitution. Nucleophile attacks carbonyl C, tetrahedral intermediate forms and leaving group departs. Amides are resonance stabilised and less reactive; hydrolysis may require acid/base and heat. Amino acids condense to peptide bonds.

Worked example: follow the reasoning

Ethanoyl chloride reacts with methylamine to give N-methylethanamide and HCl; a second amine equivalent can neutralise HCl.

Quick check · multiple choice

A What is a by-product of ethanoyl chloride reacting with ethanol?

[1]

- A HCl
- B H₂
- C O₂
- D NH₃

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Name product from ethanoyl chloride and ethanol and describe the reaction type and another product.

[3]

Answers and explanation

Quick check: A. HCl

Question 1 [3 marks]

Ethyl ethanoate [1]; nucleophilic acyl substitution/esterification [1]; HCl [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.